FLAGSHIP 07 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-TEC/D

Direct Cooling Semiconductor Vest · 2-Second Instant Solid-State Peltier Vest

2-Second Instant Peltier Cooling. Zero Water, Zero Refrigerant, 8-Hour Endurance.

Leveraging the solid-state Peltier effect via direct-contact TEC modules, the LC Frost-TEC/D requires no ice, liquid coolant, or compressor. Direct-contact thermal conduction plates absorb torso heat immediately upon power-on - stabilizing in just 2 seconds - while whisper-quiet fans dissipate waste heat. Features a breathable mesh chassis and high-visibility reflective stripes for industrial inspection, medical PPE, and outdoor operations.

01 Instant 2-Second Solid-State Conduction

Direct Peltier contact plates cool immediately upon activation with zero pre-freezing or water refilling.

02 Whisper-Quiet Vibration-Free Operation

Low-noise axial cooling fans eliminate compressor drone and pump vibrations for quiet inspections.

03 Up to 8-Hour Modular Power

Low 60W power draw enables 2-8 hours of runtime (2 hours per lightweight battery pack).

04 Zero Maintenance & 20,000-Hour Longevity

Solid-state architecture eliminates hoses, refrigerants, and pumps for leak-proof durability.

LC Frost-TEC/D · OFFICIAL HARDWARE VIEW



KEY METRIC 01
2 Seconds
Ultra-Fast Solid-State Cooling Stabilization

KEY METRIC 02
0 Liquid
Completely Dry Direct Conduction (Zero Leakage Risk)

KEY METRIC 03
2-8 Hours
Extended Modular Battery Operation

KEY METRIC 04
> 20,000 Hours
Solid-State Semiconductor Lifespan

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



4-POD SOLID-STATE PELTIER · >20,000H LIFESPAN

4-Pod Direct-Contact Solid-State Peltier Conduction Array

Four anodized aluminum thermal conduction plates coupled directly to solid-state semiconductor chips pull heat from the upper back within 2 seconds of power-on - zero water, zero pumps, and >20,000 hours of chip lifespan.



HI-VIS QUICK-RELEASE FASTENERS

Quick-Release Hi-Vis Chest Buckles & Elastic Bungee Tension

High-visibility yellow quick-release buckles and rear dynamic bungee lacing maintain optimal contact pressure between the 4 Peltier pods and the operator's upper back.



DETACHABLE WASHABLE ANTISTATIC FR MESH

Chest Rotary Controller & Modular Washable Mesh

Intuitive chest-mounted rotary dial adjusts solid-state cooling intensity on the fly, while the entire electronic harness detaches so the antistatic FR mesh vest can be washed.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **4-Pod Direct-Contact TEC Array [Quad Dorsal Solid-State Peltier Array]**
   Four Peltier conduction plates positioned on the upper back deliver instantaneous cooling directly to the skin.
2. **Quick-Release Yellow Chest Buckles [Hi-Vis Anterior Fastening Harness]**
   High-visibility chest fasteners and rear bungee cord ensure a snug, non-restrictive fit.
3. **Chest Controller & Low-Noise Fans [Rotary Thermal Dial & Axial Heat Sinks]**
   Chest-mounted circular dial provides intuitive multi-stage thermal regulation.
4. **Detachable Washable Hi-Vis Mesh [Modular Washable Antistatic Mesh Chassis]**
   Breathable antistatic mesh with neon-yellow and silver reflective stripes; electronics detach easily for washing.

FIELD DEPLOYMENT | HIGH-SPEED RAIL CORRIDOR INSPECTION | 8-HOUR PATROL SHIFT



Railway Track Inspection, Outdoor Security & Mobile Field Patrols

Mobile inspection engineers, traffic officers, and outdoor security personnel need immediate cooling without carrying water reservoirs or waiting for freezer packs. Weighing ~1.7kg and stabilizing in just 2 seconds on a 12V battery system, the LC Frost-TEC/D delivers up to 8 hours of dry, whisper-quiet solid-state cooling across long field patrols.

2 Seconds (Instant)	2-8 Hours (12V / 60W)	> 20,000 Hours
Thermal Stabilization	Modular Battery Runtime	Semiconductor Lifespan

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-TEC/D (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-TEC/D
IDENTITY & COMPLIANCE	**Color**	Tactical Grey Mesh with Neon Yellow Hi-Vis Reflective Trim & Black Straps (Black Option)
IDENTITY & COMPLIANCE	**Target Users**	Industrial inspection, medical PPE, outdoor traffic, security personnel
PERFORMANCE & MECHANICS	**Cooling Capacity**	45W
PERFORMANCE & MECHANICS	**Rated Voltage**	12V
PERFORMANCE & MECHANICS	**Rated Power**	60W
PERFORMANCE & MECHANICS	**Cooling Response Time**	Instant cooling, stable in 2s
PERFORMANCE & MECHANICS	**Battery Life**	2-8 hours (2 hours per single battery pack)
PERFORMANCE & MECHANICS	**Chip Lifespan**	> 20,000 hours
MATERIALS & ERGONOMICS	**Outer Fabric**	Antistatic and flame-retardant breathable mesh (modular detachable & washable)
DIMENSIONS & WARRANTY	**Sizes**	Standard Size / Plus Size
DIMENSIONS & WARRANTY	**Net Weight**	~1.7kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-TEC/D

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware